

ORF309/MAT380

**Bernoulli Processes IV:
Continuous Time Limit**

(pp.56-62)

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Agenda for today

- **How can we go from discrete time to continuous time**
 - Continuous Time Limit (CTL)
- **Poisson**, **Gamma**, **Exponential** distributions
- Arrivals and arrival times
 - Number of arrivals at any time
 - Time of the k -th arrival

Mitem 1 will be on Firday March 6,
9:35 – 10:15 am @ Friend 101.

It will be in-person exam.

More information will be released shortly.

**Do not forget to visit the office hours of
our two amazing UCAs on Thursdays!**

Recap → Bernoulli Process for Discrete time arrivals

- So far we have studied **Bernoulli Processes**
 - They model the **sequence of independent trials of Successes/Failures** with $P(\text{Success})=p$
 - They can be thought of as models of the arrivals of certain events
 - **Example:**
 - Frying pans manufactured in a factory:
 - Frying pan with defect (cannot be sold) → arrival of a Failure
 - Perfect frying pan (can be sold) → arrival of a Success
 - Frying pans are produced in discrete times $t \in \{10', 20', 30', \dots, 60'\}$

Continuous Time Model

Continuous Time Model

- Discrete time models cannot model every process
- **Example**: You work at an ice cream store and want to **model the customer arrival times**.
 - **Modeling Assumptions**:
 - Customers can arrive at any time $t \in [0, \infty)$
 - Arrivals are **independent** of each other
 - $P(\text{a customer arrival}) = p$ (**unknown**)
 - **Illustration**: People flip a coin every minute and if they get **Heads** they go to the ice cream store.

Coin flips are independent; **but** can people flip a coin more frequently??

Continuous Time Model

- **Example (cont.):**

- Assume we **know** the **average # of customers arriving in 1 hour** = μ
- Split 1 hour into $n = 60$ time intervals of equal length ℓ
- Define the **DRV**: $X_k = 1$, if a customer arrives at k -th interval; $X_k = 0$, otherwise
 - $X_1, X_2, X_3, \dots$ are **Bernoulli RVs** $\rightarrow P\{X_i = 1\} = p$
- **Bernoulli Process** $S_n = X_1 + X_2 + X_3 + \dots + X_n \sim \text{Binom}(n, p)$
- What is the probability of an arrival? $\rightarrow$
(we only know: μ and n)

Continuous Time Model

- **Example (cont.):**

- Assume we know the **average # of customers arriving in 1 hour = μ**
- Split 1 hour into **$n = 60$ time intervals** of equal **length ℓ**
- Define the **DRV**: $X_k = 1$, if a customer arrives at k -th interval; $X_k = 0$, otherwise
 - $X_1, X_2, X_3, \dots$ are **Bernoulli RVs** $\rightarrow P\{X_i = 1\} = p$
- **Bernoulli Process** $S_n = X_1 + X_2 + X_3 + \dots + X_n \sim \text{Binom}(n, p)$
- What is the probability of an arrival? $\rightarrow \mu = E[S_n] = np \Rightarrow p = \mu/n$
(we only know: μ and n)

continuous time scaling if $n \rightarrow \infty$

Continuous Time Model

- **Example (cont.):**

- Relationships between n (discrete time intervals) and ℓ (length) and μ ?

- $p = \frac{\mu}{n} \leq 1 \Rightarrow n \geq \mu$ and $\ell = \frac{1}{n}$ ■

- This is a good mathematical model, but it has a serious problem!
 - Why should customers arrive every 1 minute?
 - Why not every 1 second ($n = 3600$) , or 1 millisecond ($n = 3600000$), ...
 - **The Continuous Time Limit (CTL):** we can drive $n \rightarrow \infty$ and $\ell \rightarrow 0$ and be able to compute probabilities of arrivals in **continuous time** (not only discrete time)

Continuous Time Limit (CTL)

Continuous Time Limit (CTL)

- Define the **DRV**: $N = \text{"\# of customers that arrive any time within 1 hour"}$
- We will compute its distribution: $P\{N = k\} = ??$ for $k = 0, 1, 2, 3, \dots$
 - Only the **time is continuous** (customers can arrive at **any time**)
- **Recall that** for $n < \infty$ $S_n \sim \text{Binom}(n, p)$ with $p = \mu/n$
 - S_n describes customer arrivals when the **time is discrete**

(N is a discrete RV
in continuous time)

(S_n is a discrete RV,
in discrete time)

Question: how can we use the distribution of S_n to compute the distribution of N ?

Continuous Time Limit (CTL)

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 - Only the time is continuous (customers can arrive at any time)
- Recall that for $n < \infty$ $S_n \sim \text{Binom}(n, p)$ with $p = \mu/n$
 - S_n describes customer arrivals when the time is discrete

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(S_n is a discrete RV,
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- To compute $P\{N = k\}$ we will take the Continuous Time Limit of S_n

$$P\{N = k\} = \lim_{n \rightarrow \infty} P(S_n = k) = \lim_{n \rightarrow \infty} \binom{n}{k} p^k (1 - p)^{n-k} = \lim_{n \rightarrow \infty} \binom{n}{k} \left(\frac{\mu}{n}\right)^k \left(1 - \frac{\mu}{n}\right)^{n-k}$$

Continuous Time Limit (CTL)

- Let's use **Calculus** to simplify things:

$$\begin{aligned} \binom{n}{k} \left(\frac{\mu}{n}\right)^k \left(1 - \frac{\mu}{n}\right)^{n-k} &= \frac{n!}{k! (n-k)!} \frac{\mu^k}{n^k} \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-k} = \\ &= \frac{\mu^k}{k!} \frac{n(n-1) \dots (n-k+1)}{n^k} \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-k} = \frac{\mu^k}{k!} \frac{(n-1) \dots (n-k+1)}{n^{k-1}} \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-k} \end{aligned}$$

But: $\lim_{n \rightarrow \infty} \frac{(n-1) \dots (n-k+1)}{n^{k-1}} = 1$, $\lim_{n \rightarrow \infty} \left(1 - \frac{\mu}{n}\right)^n = e^{-\mu}$, $\lim_{n \rightarrow \infty} \left(1 - \frac{\mu}{n}\right)^{-k} = 1$

Refresh your Calculus and prove this!

$$\text{Hence: } \mathbf{P}\{N = k\} = \lim_{n \rightarrow \infty} \mathbf{P}(S_n = k) = \frac{\mu^k}{k!} e^{-\mu}$$

Poisson distribution

- **Definition:** A DRV Z with values in $\{0,1,2, \dots\}$ such that

$$P\{Z = k\} = \frac{e^{-\mu} \mu^k}{k!}, \quad 0 \leq k < \infty$$

is said to have the **Poisson distribution** with mean μ

- **Notation:** $Z \sim \text{Pois}(\mu)$
- **Remark:** The Poisson distribution provides the probability that k customers arrive **at any time within one time unit** when the **rate of arrivals** is known to be μ
- **Remark:** The **Poisson distribution** was obtained by taking the **CTL** of the **Binomial distribution**.

See Exercise 1 at the end of the slides!

Poisson distribution

- **Sanity checks** (is it indeed a probability distribution?):

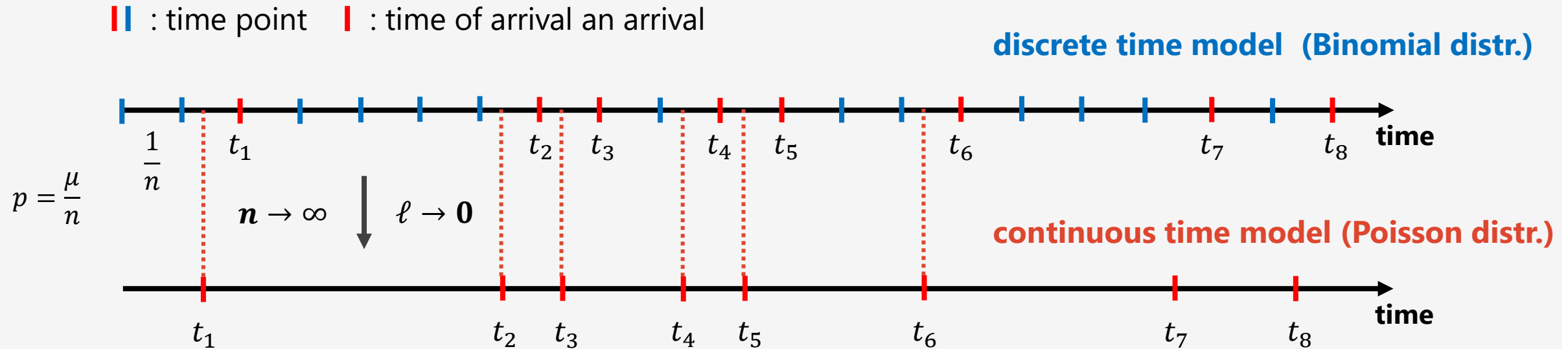
$$\sum_{k=0}^{\infty} P\{N = k\} = \sum_{k=0}^{\infty} \frac{e^{-\mu} \mu^k}{k!} = e^{-\mu} \sum_{k=0}^{\infty} \frac{\mu^k}{k!} = e^{-\mu} e^{\mu} = 1$$

Taylor expansion
of e^{μ} at 0

$$E[N] = \sum_{k=0}^{\infty} k P\{N = k\} = e^{-\mu} \sum_{k=0}^{\infty} k \frac{\mu^k}{k!} = e^{-\mu} \mu \sum_{k=1}^{\infty} \frac{\mu^{k-1}}{(k-1)!} = e^{-\mu} \mu e^{\mu} = \mu \quad \blacksquare$$

Taylor expansion
of e^{μ} at 0

Illustration of CTL



- t_i is the **arrival time** of the i -th customer
 - In the **discrete time model** customers can arrive at specific discrete time points $t_i \in \{1, 2, 3, \dots\}$
 - In the **continuous time model** customers can arrive at any time $t_i \in \mathbb{R}_+$
- By taking $n \rightarrow \infty$ we force $\ell \rightarrow 0$ and we **convert** the discrete time into continuous time

Time of arrivals in t time units

- So far, we have considered the average number of arrivals **within only 1 hour**
(or 1 time unit)
 - The average number of arrivals is known in 1 hour: μ
- What happens if we have: 2 or 3.52 hours, or 0.5 or 0.123 hours?
- Define the **DRV**: $N_t =$ “the number of arrivals **by time t** ” where $t \in [0, \infty)$
 - Then the **average** number of arrivals will be ??

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 - The average number of arrivals is known in 1 hour: μ
- What happens if we have: 2 or 3.52 hours, or 0.5 or 0.123 hours?
- Define the **DRV**: $N_t = \text{"the number of arrivals by time } t\text{"}$ where $t \in [0, \infty)$
 - Then the **average** number of arrivals will be μt
- Working similarly as before we can derive: $N_t \sim \text{Pois}(\mu t)$
- **Note:** $N_1 = N$, (i.e., the Poisson RV we discussed before.)

Example: Flash Crashes in High Frequency Trading

A **flash crash** is a sudden jump (**up** or **down**) of a stock price in a trading day.

These are rare events, but they have a **significant impact on trading algorithms** and **cause large losses**

Example:

The rare event: a jump of more than 3% in the price of a security within 1 trading day

Historical data: we can estimate that it happens **once every 5 days** (trading week)

Objective: we want to determine the probability of such events happening in a **single day**

All data are imaginary and for educational purposes!

We can determine the **average rate** of a flash crash in a single day: $\mu = \frac{1}{5} = 0.2$

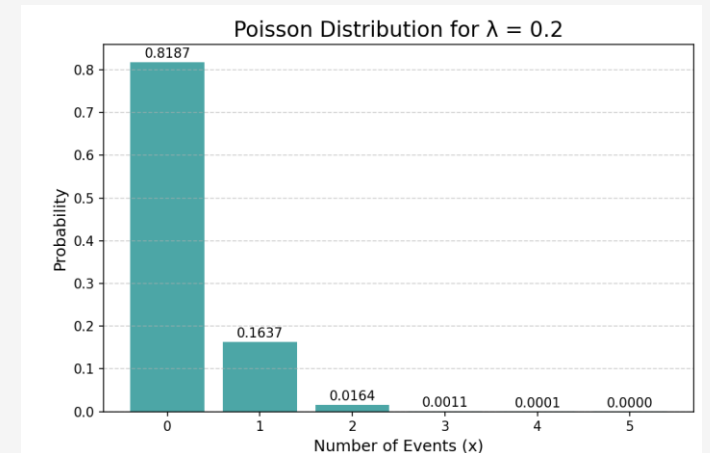
$N = \text{"# of flash crashes in a day"} \rightarrow N \sim \text{Pois}(\mu = 0.2)$

❖ Here are some probabilities of possible flash crashes:

$$P\{N = 0\} = \frac{0.2^0}{0!} e^{-0.2} \approx 0.8187$$

$$P\{N \geq 1\} = 1 - P\{N = 0\} = 1 - 0.8187 \approx 0.1813$$

$$P\{N = 5\} = \frac{0.2^5}{5!} e^{-0.2} \approx 3.25 \times 10^{-6}$$



Example: Flash Crashes in High Frequency Trading

Some historical Flash Crashes:

1. The **Panic of 1890 (Baring Crisis)**: It caused a short severe recession in U.S. that was sparked by the potential collapse of London's Barings Bank following failed investments in Argentina.
2. The **2010 Flash Crash**: it occurred on May 6, 2010.
The **Dow Jones Industrial Average** plummeted over **1000 points in minutes**, which represented a **plunge of over 9%**, and briefly wiped out approximately **\$1 trillion** in market value. **By the end of the day most of the losses were recovered.**
* Accenture's price went from **\$38 at 2:47:43pm** to **\$0.01 at 2:47:54pm** to **\$41 at 3:00:00pm**
3. The **2016 British Pound Flash Crash**: it occurred on October 6, 2016, causing the British pound to fall dramatically against the U.S. dollar in a short period.

Historical information obtained using GAI

Time of the k -th arrival

Time of the k -th arrival

- **Question:** What is the probability that the k -th customer **arrives by time t** ?
- Define the RV: T_k = “the time of the k -th arrival” T_k is a continuous RV, because its values are in $[0, \infty)$
- We are looking for the **distribution** of T_k , that is: $P\{T_k \leq t\}$ for all $t \in [0, \infty)$
- How can I write $\{T_k \leq t\}$ in terms of N_t ??

Time of the k -th arrival

- **Question:** What is the probability that the k -th customer **arrives by time t** ?
- Define the RV: T_k = “the time of the k -th arrival” T_k is a continuous RV, because its values are in $[0, \infty)$
- We are looking for the **distribution** of T_k , that is: $P\{T_k \leq t\}$ for all $t \in [0, \infty)$
- How can I write $\{T_k \leq t\}$ in terms of N_t $\rightarrow \{T_k \leq t\} = \{N_t \geq k\}$
- Because it is **equivalent** to say:
 - The k -th customer arrived by time t
(e.g., the k -th customer may arrive at time $t - \epsilon$ and another customer may arrive at time t)
 - The number of arrivals by time t is **at least** k

Time of the k -th arrival

- Hence:

$$P\{T_k \leq t\} = P\{N_t \geq k\} = 1 - P\{N_t < k\} = 1 - \sum_{i=0}^{k-1} \frac{e^{-\mu t} (\mu t)^i}{i!}$$

- **Note:** we cannot simplify this distribution further! But that's OK, computers have no problem computing it efficiently!

Gamma distribution

- **Definition:** A random variable T with values in $[0, \infty)$ such that

$$P\{T \leq t\} = 1 - \sum_{i=0}^{k-1} \frac{e^{-\mu t} (\mu t)^i}{i!}, \quad t \geq 0$$

is said to have the **Gamma distribution** with shape k and scale μ

- **Notation:** $T \sim \text{Gamma}(k, \mu)$

Time of the 1-st arrival

Time of the 1st arrival

- The time of the 1st arrival is always very important (e.g., 1st customer, 1st bus, ...)

$$P\{T_1 \leq t\} = 1 - \sum_{i=0}^{1-1} \frac{e^{-\mu t} (\mu t)^i}{i!} = 1 - e^{-\mu t}$$

- **Definition:** A random variable T with values in $[0, \infty)$ such that

$$P\{T \leq t\} = 1 - e^{-\mu t}$$

is said to have the **Exponential distribution** with parameter μ

- **Notation:** $T \sim \text{Expon}(\mu)$

Time of the *1st* arrival *within an interval*

- **Example:** Compute the probability that the 1st arrival will come in the interval $(a, b]$??
 - We are looking for the probability of the event: $\{a < T_1 \leq b\}$??

Time of the 1st arrival *within an interval*

- **Example:** Compute the probability that the 1st arrival will come in the interval $(a, b]$??
 - We are looking for the probability of the event: $\{a < T_1 \leq b\}$??
 - We can write: $\{T_1 \leq b\} = \{T_1 \leq a\} \cup \{a < T_1 \leq b\}$

$$P\{T_1 \leq b\} = P\{T_1 \leq a\} + P\{a < T_1 \leq b\} \Rightarrow$$

$$\Rightarrow P\{a < T_1 \leq b\} = P\{T_1 \leq b\} - P\{T_1 \leq a\} = 1 - e^{-\mu b} - 1 + e^{-\mu a}$$

$$\Rightarrow P\{a < T_1 \leq b\} = e^{-\mu a} - e^{-\mu b}$$

Time of the 1st arrival at a *precise time point* t

- **Example:** Compute the probability $P\{T_1 = t\} = ??$
 - **Note:** T_1 is a continuous RV and we as we stated: $P\{T_1 = t\} = 0$
 - We can verify this using the **CTL** and considering the event: $\{t - \epsilon < T_1 \leq t\}$

$$P\{T_1 = t\} = \lim_{\epsilon \rightarrow 0} P\{t - \epsilon < T_1 \leq t\} = \lim_{\epsilon \rightarrow 0} (e^{-\mu(t-\epsilon)} - e^{-\mu t}) = 0$$

- **Remark:** The formula for the expectation of **discrete RVs** cannot be applied on **continuous RVs**, as it would have suggested that the expected arrival time of 1st customer is 0:

$$E[T_1] = \sum_{t \in D} t P\{T_1 = t\} = 0 \quad \leftarrow \text{NOT TRUE!}$$

Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

Exercise 1 (Example)

- In a manufacturing facility in which glass items are produced, defects occur occasionally, making the item unsuitable for selling it to customers. Historically, 1 in every 1000 of the produced items has a defect. If we inspect a random sample of 8000 items, what is the probability that fewer than 7 items have some defect?
- **Solution:** This exercise offers a very good example to show the Binomial distribution converges to the Poisson distribution when the number of trials is very large. (before you see the solution try to solve it yourselves! You should model this problem using a Binomial RV and a Poisson RV) → → →

-
- We first solve the problem defining a DRV $S_n = \text{"\# defects in } n \text{ trials"}$, $n = 8000$
 - Obviously, a defect can be thought of as a failure, with probability $p = 0.001$
 - Hence: $S_n \sim \text{Binom}(p = 0.001, n = 8000)$
 - We are interested in computing:

$$P\{S_n < 7\} = P\{S_n \leq 6\} = \sum_{k=1}^6 \binom{8000}{6} 0.001^k 0.999^{8000-k} = \dots$$

Good luck with the computations
and the numerical errors...

- It is very *messy* to compute this summation as it involves very large numbers

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- Let's solve it using the Poisson distribution. We need the rate by which the defects come.
 - Since $n = 8000 \gg 0$ we can consider the rate: $\lambda = np = 8000 \times 0.001 = 8$
 - Define the DRV: $X = \text{"# of defects in a sample"}$. Then $X \sim \text{Pois}(\lambda = 8)$

$$P\{X < 7\} = P\{X \leq 6\} = \sum_{k=1}^6 e^{-8} \frac{8^k}{k!} \approx 0.31$$

- Note: this summation is more convenient to compute when we have large numbers.

Exercise 2 (Example)

- Consider the Poisson RV: $X \sim \text{Pois}(\lambda)$. Show the following

$$(i) \ E[X(X-1) \dots (X-(m-1))] = \lambda^m, \quad (ii) \ \text{Var}(X) = \lambda$$

- Solution:**

$$\begin{aligned} (i) \ E[X(X-1) \dots (X-(m-1))] &= \sum_{x=0}^{\infty} x(x-1) \dots (x-m+1) e^{-\lambda} \frac{\lambda^x}{x!} = \\ &= \sum_{x=0}^{\infty} x(x-1) \dots (x-m+1) e^{-\lambda} \frac{\lambda^x}{[x(x-1) \dots (x-m+1)](x-m)!} = \\ &= e^{-\lambda} \sum_{x=m}^{\infty} \frac{\lambda^x}{(x-m)!} = e^{-\lambda} \lambda^m \sum_{x=m}^{\infty} \frac{\lambda^{x-m}}{(x-m)!} = e^{-\lambda} \lambda^m e^{\lambda} = \lambda^m \end{aligned}$$

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- To show (ii) we will use $E[X] = \lambda$ and the result we obtained in (i), and the definition of the variance: $Var(X) = E[X^2] - (E[X])^2$ (1)
 - More specifically, note that: $E[X^2] = E[X(X - 1)] + E[X]$ (2)
 - To use (i) we write: $E[X(X - 1)] = E[X(X - (2 - 1))]$. Hence $m = 2$
 - Then we have $E[X(X - 1)] = \lambda^2$ (3)
 - Substituting (2) and (3) into (1) yields:

$$Var(X) = E[X(X - 1)] + E[X] - (E[X])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$



Exercise 3

- In the old-days, at a busy time a telephone exchange is very near capacity, so people could not find a line to use. It was very important to know the number of attempts necessary in order to gain a connection. Suppose that the probability of a connection during busy times is $p = 0.05$. What is the probability that 5 attempts are necessary for a successful connection.

Exercise 4

- The average number of oil tankers arriving at a certain port city is known to be 10. the facilities at the port can handle at most 15 tankers a day. What is the probability that on a given day tankers will have to be sent away?
